

NYC Yellow Taxi Data Pipeline

End-to-end batch data pipeline for NYC Yellow Taxi trip data, built on Databricks using PySpark, Delta Lake, and a medallion (Bronze / Silver / Gold) architecture. Orchestrated as a multi-task Databricks Job and visualized in Power BI.

Architecture

Raw data → **Bronze** (raw ingestion) → **Silver** (cleaned + quarantined) → **Gold** (business aggregates) → **Power BI dashboard**

Bronze — Ingests the monthly NYC TLC Yellow Taxi trip data (parquet) and taxi zone lookup (CSV) from the official TLC source, landing them as Delta tables with ingestion metadata (`_ingested_at`, `_source_file`). Files are staged through a Unity Catalog Volume, since Databricks Free Edition's serverless compute blocks direct local-filesystem access from Spark. Reloads are idempotent — re-running a load for the same month deletes the prior copy first, rather than duplicating it.

Silver — Applies explicit data quality rules. Rows with an invalid fare, distance, duration, or an unmapped pickup/dropoff zone are hard-rejected into a separate `silver_yellow_taxi_rejected` table with a reason code per row. Unreliable `passenger_count` values are soft-corrected (nulled) rather than discarding the whole trip. Clean trips are enriched with pickup/dropoff borough and zone names.

Gold — Five pre-aggregated tables, each answering one specific business question: daily trend, hourly demand patterns, busiest zones, payment mix, and borough-to-borough flow.

Orchestration — A 3-task Databricks Job (Bronze → Silver → Gold) with dependency-based execution, parameterized by month (`year_month`) to support backfilling additional months.

Visualization — Power BI dashboard built from the Gold tables (KPI cards, daily trend line, hour/day heatmap, top zones, payment mix, borough flow matrix).

Results (May 2026 data)

4,090,836 raw trip records processed

4.4% rejection rate (179,015 rows quarantined with an explicit reason each)

5 Gold aggregate tables feeding a 6-visual Power BI dashboard

Notebooks

Notebook	Purpose
<code>01_bronze_ingestion.py</code>	Downloads and lands raw trip + zone data
<code>02_silver_cleaning.py</code>	Cleans, quarantines, and enriches trip data
<code>03_gold_aggregation.py</code>	Builds the 5 business-ready aggregate tables

Documentation

See the included explainer document for a full plain-English, line-by-line walkthrough of every piece of code in this pipeline.

Dashboard

See the `screenshots` folder for the final Power BI dashboard.

Tools

Databricks, PySpark, Delta Lake, Unity Catalog Volumes, Databricks Workflows, Power BI Desktop.